VILLA COMPLEX 415V LV NETWORK

# Electrical design report

| | |
|---|---|
| Project | Villa Complex 415V LV Network |
| Project no. | *Not defined* |
| Client | Mr. Ahmed Al Mansoori |
| Consultant | ABC Engineering Consultants |
| Contractor | *Not defined* |
| Plot / area | 123-456 · VILLA, UAE |
| Scope | Whole installation (5 boards) |
| Studies | Load assessment; Transformer and standby generator sizing; LV distribution; Cable and breaker sizing; Load flow and voltage drop study; Short circuit study; Protection discrimination study; Earth fault loop impedance and disconnection; Phase balance; Busbar trunking risers; Power factor correction; DB and cable schedules |
| System | 415 V, 3-phase + N, 50 Hz, TN-S |
| Document no. | E-RPT-001 |
| Revision | *Not defined* |
| Date | 15 October 2026 |
| Issue status | *Not defined* |

| | Name | Signature | Date |
|---|---|---|---|
| **Prepared by** | QA | | |
| **Checked by** | *Not defined* | | |
| **Approved by** | *Not defined* | | |

Generated by LV Design Studio from the latest calculation run. Calculations cover the whole network; this report shows the part in scope. Calculated checks against the project limits — engineering review and approval are separate. Results must be checked by a qualified engineer before submission. **Cable data — 2.5, 4, 6, 10, 16, 25, 35, 50, 95, 150, 300 mm²: app reference value (approximate, not manufacturer-verified). Verify against the manufacturer's datasheet before relying on these results.**

## DOCUMENT CONTROL

### Document information

| | |
|---|---|
| Document title | Electrical design report |
| Document no. | E-RPT-001 |
| Project | Villa Complex 415V LV Network |
| Project no. | *Not defined* |
| Current revision | *Not defined* |
| Issue status | *Not defined* |
| Date of this print | 15 October 2026 |

### Revision history

> No revision has been issued for this project (Revisions → Issue revision). This report is an unissued draft.

### Prepared, checked and approved

| Role | Name | Signature | Date |
|---|---|---|---|
| **Prepared by** | QA | | |
| **Checked by** | *Not defined* | | |
| **Approved by** | *Not defined* | | |

## CALCULATION RECORD

*What these results were calculated from, with what, and how far they can be relied on.*

| | |
|---|---|
| Project | Villa Complex 415V LV Network<br>Project ID not assigned (save the project) · file format version 1 |
| Calculated | At the time of this print, from the current inputs |
| Software | LV Design Studio 1.38.0 · built-in calculation engine (no external engine used) |
| Calculation basis | Methods per Codes and standards → Calculation basis and each study's Method notes.<br>Standard editions: not verified against a specific edition |
| Key assumptions | • 415 V, 3-phase + N, 50 Hz, TN-S · design ambient 45 °C · voltage drop limit 0.5 % source to load<br>• Cable resistance at operating temperature 1.2 × R20 (not set for the project)<br>• Fault levels: maximum Ik″ with c max = 1.05 and cable R at 20 °C, MV network infinite, transformer correction KT not applied — **simplified, preliminary, not a full IEC 60909 study** |
| Cable data | **11 sizes used: 11 app reference values (approximate, not manufacturer-verified) · 0 manufacturer data**<br>Final-design requirement for manufacturer cable data: not set |
| Input checks | No invalid inputs, no supply loops, no missing references |
| Calculation warnings | 49 failures · 36 to check (see the Results summary under Contents) |
| Limitations | • Cable ratings: per installation method with ambient / ground, soil and grouping factors; grouping with other circuits is counted only on cable tray routes<br>• Selectivity from current thresholds only — confirm with the manufacturer's tables<br>• Voltage drop: first-order, summed along the radial path; not an unbalanced load flow |

**Validation status: PRELIMINARY.** Calculated checks against the project limits, using approximate reference or unverified cable data, with failing checks. Not an engineering approval and not an approval by DEWA or any other authority.

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### Results summary

| Section | Study | Result |
|---|---|---|
| 6 | Load assessment | — |
| 7 | Transformer and standby generator sizing | 2 pass · 0 warning · 0 fail |
| 8 | LV distribution | 9 pass · 0 warning · 0 fail |
| 9 | Cable and breaker sizing | **0 pass · 1 warning · 35 fail** |
| 10 | Load flow and voltage drop study | **7 pass · 1 warning · 14 fail** |
| 11 | Short circuit study | 36 pass · 0 warning · 0 fail |
| 12 | Protection discrimination study | **0 pass · 31 warning · 0 fail** |
| 13 | Earth fault loop impedance and disconnection | 36 pass · 0 warning · 0 fail |

| Section | Study | Result |
|---|---|---|
| 14 | Phase balance | **3 pass · 2 warning · 0 fail** |
| 15 | Busbar trunking risers | — |
| 16 | Power factor correction | **0 pass · 1 warning · 0 fail** |
| 17 | DB and cable schedules | — |

## 1. EXECUTIVE SUMMARY

| Connected load | Maximum demand | Demand ÷ connected | Power factor · current |
|---|---|---|---|
| **514 kW** | **451 kW · 512 kVA** | **0.88** | **0.88 · 712 A** |

### 1.1 Transformer requirement

**Table 1**

| Main board | Required (kVA) | Recommended | Installed (kVA) | Loading | Status |
|---|---|---|---|---|---|
| MDB-1 | 710 | 1000 kVA | 1,000 | 47 % | PASS |

### 1.2 Generator requirement

**Table 2**

| Essential demand (kVA) | Rating needed (kVA) | Governed by | Recommended | Installed (kVA) | Status |
|---|---|---|---|---|---|
| 64 | 544 | motor start | 630 kVA / 504 kW | *Not defined* | *NOT CHECKED* |

### 1.3 Main LV distribution

- MDB-1 (MDB) feeding 4 board(s): 2 SMDB, 1 DB, 1 MCC

### 1.4 Major design findings

**Table 3**

| Status | Finding |
|---|---|
| PASS | Transformer and standby generator sizing: all 2 checks pass. |
| PASS | LV distribution: all 9 checks pass. |
| **FAIL** | Cable and breaker sizing: 35 check(s) fail. |
| **FAIL** | Load flow and voltage drop study: 14 check(s) fail. |
| PASS | Short circuit study: all 36 checks pass. |
| **WARNING** | Protection discrimination study: 31 check(s) need review. |
| PASS | Earth fault loop impedance and disconnection: all 36 checks pass. |
| **WARNING** | Phase balance: 2 check(s) need review. |
| **WARNING** | Power factor correction: 1 check(s) need review. |

*Whole installation, from the latest calculation run. Section results are for the scope of this report.*

## 2. PROJECT SCOPE

> The project scope is not defined (Project → Brief). Included and excluded systems are not listed.

### 2.1 Electrical systems included

*Not defined*

### 2.2 Electrical systems excluded

*Not defined*

### 2.3 Design boundaries

- Starts at: the LV terminals of the transformer feeding MDB-1.
- MV network upstream of the transformers is outside the calculations (treated as an infinite source).
- Ends at: the final circuits on the DB load schedules and the loads connected to them.

## 3. ELECTRICAL SYSTEM DESCRIPTION

### 3.1 Incoming supply

- MDB-1: from a dedicated transformer

### 3.2 MV / LV arrangement and transformers

- MDB-1: 1000 kVA, 5 % Z, Dyn11 (app default), 415 V secondary

### 3.3 Main LV distribution

- 415 V, 3-phase + N, 50 Hz, TN-S (the earthing arrangement the calculations assume).
- MDB-1 (MDB) feeding 4 board(s): 2 SMDB, 1 DB, 1 MCC

### 3.4 Emergency supply

None defined in the project.

### 3.5 UPS

None defined in the project.

# 4. CODES AND STANDARDS

## 4.1 Project and authority requirements

| | |
|---|---|
| Approving authority | *Not defined* |

> No project standards or specifications are defined (Reports → Study reports → Output). None are assumed.

## 4.2 Calculation basis

**Table 4**

| Standard | Used for |
|---|---|
| IEC 60909-0 | Short-circuit currents (simplified: c factor, infinite MV network) |
| IEC 60364-5-52 | Cable current-carrying capacity, derating and voltage drop |
| IEC 60364-4-43 | Overload protection: Ib ≤ In ≤ Iz |
| IEC 60364-4-41 | Automatic disconnection: earth fault loop impedance and disconnection times |
| IEC 60364-5-54 | Protective conductor size (Table 54.2 / adiabatic) |
| IEC 60898-1 / IEC 60947-2 | Circuit-breaker characteristics and breaking capacity |
| IEC 60831 | Capacitor bank rating and switching device |
| IEC 61439-6 | Busbar trunking ratings |

*The methods the app's calculations follow for the studies in this report. They are not a statement of the project's contractual standards.*

## 5. DESIGN CRITERIA

**Table 5**

| Parameter | Design value | Source | Note |
|---|---|---|---|
| System voltage (line-line) | 415 V | Project settings | |
| Frequency | 50 Hz | Project settings | |
| Earthing arrangement | TN-S | Calculation basis (fixed) | |
| Design ambient temperature | 45 °C | Project settings | |
| Voltage drop limit (source to load) | 0.5 % | Project settings | |
| Conductor temperature for voltage drop | 1.2 × R20 | Calculation basis | *Not set for the project* |
| Final circuit disconnection | IEC 60364-4-41 Table 41.1 | Project settings | |
| Power factor target | 0.95 | App default | *Not set for the project — app default used* |
| Future growth allowance (transformer) | 20 % | App default | *Not set for the project — app default used* |
| Transformer maximum design loading | 80 % | App default | *Not set for the project — app default used* |
| Generator maximum design loading | 80 % | App default | *Not set for the project — app default used* |
| Maximum demand factor (authority forms) | 0.8 | Project information | |
| Overall demand ÷ connected load | 0.88 (451 ÷ 514 kW) | Load assessment (calculated) | |

## 6. LOAD ASSESSMENT

Scope: Whole installation (5 boards)

### 6.1 Input

| | |
|---|---|
| **Loads** | Connected loads on the feeders and DB load schedules |
| **Demand factors** | Per load / circuit as entered |

### 6.2 Design criteria

| | |
|---|---|
| **Future growth allowance** | 20 % (app default) |
| **Transformer design loading** | 80 % |

### 6.3 Method

- Demand = connected load × demand factor for each load, summed up the network (kW and kvar separately); maximum demand kVA = √(kW² + kvar²).
- Required capacity = maximum demand × (1 + growth) ÷ design loading, from the transformer sizing (no further diversity applied).

### 6.4 Result

| Connected load | Maximum demand | Demand ÷ connected |
|---|---|---|
| **514 kW** | **451 kW** | **0.88** |

### 6.5 Load at each board

**Table 6**

| Board | Connected (kW) | Demand (kW) | Demand ÷ connected | Demand (kVA) | PF | Current (A) |
|---|---|---|---|---|---|---|
| MDB-1 | 514 | 451.1 | 0.88 | 512.1 | 0.88 | 712 |
| SMDB-GF | 231 | 203.3 | 0.88 | 233.7 | 0.87 | 325 |
| DB-GF1 | 38 | 38 | 1 | 42.2 | 0.9 | 59 |
| SMDB-FF | 109 | 87 | 0.8 | 97.4 | 0.89 | 135 |
| MCC-1 | 130 | 130 | 1 | 151.2 | 0.86 | 210 |

### 6.6 Design load of each main board

**Table 7**

| Main board | Maximum demand (kVA) | Future growth | Required capacity (kVA) |
|---|---|---|---|
| MDB-1 | 473 | 20 % | 710 |

## 7. TRANSFORMER AND STANDBY GENERATOR SIZING

Scope: Whole installation (5 boards)

Overall status: **PASS** (2 pass · 0 warning · 0 fail)

### 7.1 Design criteria

| | |
|---|---|
| **Future growth** | 20 % |
| **Transformer design loading** | ≤ 80 % |
| **Generator design loading** | ≤ 80 % |

### 7.2 Method

- Transformer per main board: maximum demand (after the planned power factor correction) × (1 + 20 % growth) ÷ 80 % design loading, next DEWA standard size (500 / 1000 / 1500 kVA).
- For that size: full-load current and main breaker, LV fault level (typical IEC 60076-5 impedance, infinite MV source), voltage regulation at the design demand.
- Bus couplers: with one transformer out, the other carries both boards up to 100 % of its rating. Duty / standby: two transformers, each for the whole load.
- Standby generator from the boards on it (share of each), plus circuits marked essential. The set must carry the running kVA and the running kW at 80 % loading: rating ≥ max(kVA ÷ 0.8, kW ÷ (0.8 × 0.8)), with rated kW = 0.8 × rated kVA unless the Generators list gives the set's kW; or larger if the largest motor, started last, would dip the voltage over 15 % (X′d 25 %).

### 7.3 Result

| Items checked | MDB-1 transformer | Standby generator |
|---|---|---|
| **2 pass · 0 warning · 0 fail** | **1000 kVA** | **630 kVA / 504 kW** |

### 7.4 Transformers

**Table 8**

| Main board | Demand (kVA) | PF | Design (kVA) | Recommended | Installed · loading | FLC · main breaker | LV fault (kA) · lowest Icu | Regulation | Result |
|---|---|---|---|---|---|---|---|---|---|
| MDB-1 | 473 | 0.95 | 710 | 1000 kVA | 1000 kVA · 47 % | 1391 A · 1600 A | 29.2 · 36 | 1.1 % | PASS |

### 7.5 Standby generator — size

**Table 9**

| Item | Value |
|---|---|
| Running demand | 55 kW · 64 kVA |
| For the running load | 86 kVA — running; the motor start sets the size |
| Largest motor start | MCC-FP: 384 kVA starting with 0 kVA already running |
| For the motor start | 544 kVA (dip 13.2 % on the recommended set) |

| Item | Value |
|---|---|
| Recommended | 630 kVA / 504 kW at 0.8 PF |
| Result | PASS |
| With a soft starter on the largest motor | 300 kVA |
| Full-load current · ATS / breaker | 876 A · 1000 A |
| Installed | — |

### 7.6 Verification — Capacity verification

**Table 10**

| Equipment | Calculated demand (kVA) | Required capacity (kVA) | Selected | Installed (kVA) | Loading | Spare (kVA) | Status |
|---|---|---|---|---|---|---|---|
| Transformer MDB-1 | 473 | 710 | 1000 kVA | 1000 | 47 % | 527 | PASS |
| Standby generator | 64 | 544 | 630 kVA | — | — | — | *NOT CHECKED* |

*See also Appendix D (equipment data).*

## 8. LV DISTRIBUTION

Scope: Whole installation (5 boards)

Overall status: **PASS** (9 pass · 0 warning · 0 fail)

### 8.1 Input

| | |
|---|---|
| **Supply** | MDB-1: transformer 1000 kVA, 5 % Z |

### 8.2 Design criteria

| | |
|---|---|
| **Board loading** | Demand current ≤ busbar / main device rating |
| **Feeders** | Ib ≤ In ≤ Iz |

### 8.3 Method

- Board demand from the load flow; loading = demand current ÷ the board's rated current (when entered).

### 8.4 Result

| Boards | Major feeders | Checks |
|---|---|---|
| **5** | **4** | **9 / 0 / 0 (pass / check / fail)** |

### 8.5 Distribution boards

**Table 11**

| Board | Type | Fed from | Bus rating (A) | Demand (A) | Loading | Ik″ (kA) |
|---|---|---|---|---|---|---|
| MDB-1 | MDB | Supply | 1600 | 712 | 45 % | 29.2 |
| SMDB-GF | SMDB | MDB-1 | 630 | 325 | 52 % | 21.8 |
| DB-GF1 | DB | SMDB-GF | 100 | 59 | 59 % | 10.7 |
| SMDB-FF | SMDB | MDB-1 | 250 | 135 | 54 % | 14.3 |
| MCC-1 | MCC | MDB-1 | 400 | 210 | 53 % | 19 |

### 8.6 Major feeders

**Table 12**

| From | To | Breaker | Cable | Ib (A) | Iz (A) | Status |
|---|---|---|---|---|---|---|
| MDB-1 | SMDB-GF | 500 A MCCB | 4C × 300 mm² | 327 | 525 | PASS |

| From | To | Breaker | Cable | Ib (A) | Iz (A) | Status |
|---|---|---|---|---|---|---|
| MDB-1 | SMDB-FF | 200 A MCCB | 4C × 95 mm² | 135 | 266 | PASS |
| MDB-1 | MCC-1 | 315 A MCCB | 4C × 150 mm² | 210 | 342 | PASS |
| SMDB-GF | DB-GF1 | 100 A MCCB | 4C × 35 mm² | 60 | 146 | PASS |

## 9. CABLE AND BREAKER SIZING

Scope: Whole installation (5 boards)

Overall status: **FAIL** (0 pass · 1 warning · 35 fail)

### 9.1 Input

| | |
|---|---|
| **Cable** | Cu / XLPE multicore, rated for each circuit's installation method (IEC 60364-5-52) |
| **Ambient temperature** | 45 °C |
| **Ground (buried cables)** | 20 °C, soil 2.5 K·m/W (IEC reference where not set) |
| **Grouping** | From the cable tray route (method E), else the circuit's own parallel runs |

### 9.2 Design criteria

| | |
|---|---|
| **Overload protection** | Ib ≤ In ≤ Iz (IEC 60364-4-43) |
| **Voltage drop limit** | 0.5 % |
| **Breaking capacity** | Icu ≥ Ik″ at the breaker |

### 9.3 Method

- Cu/XLPE/SWA multicore cable, rated for each circuit's IEC 60364-5-52 installation method (E free air / tray unless set); derated for air ambient (methods A–E) or ground temperature and soil thermal resistivity (D1 / D2, reference 20 °C and 2.5 K·m/W), and for grouping (the cable tray route for method E, else the circuit's own parallel runs: bunched for A–C, in ducts / in the ground for D1 / D2).
- Overload protection per IEC 60364-4-43: Ib ≤ In ≤ Iz. Voltage drop from the main board to the load within the limit. Breaker Icu ≥ the fault at its terminals.

### 9.4 Result

| Circuits checked | Ambient |
|---|---|
| **0 pass · 1 warning · 35 fail** | **45 °C** |

### 9.5 Selection — major feeders

**Feeder: MDB-1 → SMDB-GF**

| | |
|---|---|
| Design current (Ib) | 327 A |
| Protective device (In) | 500 A MCCB |
| Selected cable | 4C × 300 mm² |
| Installation | E · derating 0.87 · 1.00 |
| Derated capacity (Iz) | 525 A |
| Ib ≤ In ≤ Iz | PASS |
| Voltage drop (total) | 0.46 % (limit 0.5 %) |
| Breaking capacity check | PASS |
| Overall status | **WARNING** |

**Feeder: MDB-1 → SMDB-FF**

| | |
|---|---|
| Design current (Ib) | 135 A |
| Protective device (In) | 200 A MCCB |
| Selected cable | 4C × 95 mm² |
| Installation | E · derating 0.87 · 1.00 |
| Derated capacity (Iz) | 266 A |
| Ib ≤ In ≤ Iz | PASS |
| Voltage drop (total) | 0.75 % (limit 0.5 %) |
| Breaking capacity check | PASS |
| Overall status | **FAIL** |

**Feeder: MDB-1 → MCC-1**

| | |
|---|---|
| Design current (Ib) | 210 A |
| Protective device (In) | 315 A MCCB |
| Selected cable | 4C × 150 mm² |
| Installation | E · derating 0.87 · 1.00 |
| Derated capacity (Iz) | 342 A |
| Ib ≤ In ≤ Iz | PASS |
| Voltage drop (total) | 0.58 % (limit 0.5 %) |
| Breaking capacity check | PASS |
| Overall status | **FAIL** |

**Feeder: SMDB-GF → DB-GF1**

| | |
|---|---|
| Design current (Ib) | 60 A |
| Protective device (In) | 100 A MCCB |
| Selected cable | 4C × 35 mm² |
| Installation | E · derating 0.87 · 1.00 |
| Derated capacity (Iz) | 146 A |
| Ib ≤ In ≤ Iz | PASS |
| Voltage drop (total) | 0.92 % (limit 0.5 %) |
| Breaking capacity check | PASS |
| Overall status | **FAIL** |

### 9.6 Verification — Circuits on each board

**Table 13**

| Board | Circuits | Pass / check / fail | Status |
|---|---|---|---|
| MDB-1 | 5 | 0 / 1 / 4 | **FAIL** |
| SMDB-GF | 4 | 0 / 0 / 4 | **FAIL** |

| Board | Circuits | Pass / check / fail | Status |
|---|---|---|---|
| DB-GF1 | 21 | 0 / 0 / 21 | **FAIL** |
| SMDB-FF | 4 | 0 / 0 / 4 | **FAIL** |
| MCC-1 | 2 | 0 / 0 / 2 | **FAIL** |

*See also Appendix B (cable calculations).*

## 10. LOAD FLOW AND VOLTAGE DROP STUDY

Scope: Whole installation (5 boards)

Overall status: **FAIL** (7 pass · 1 warning · 14 fail)

### 10.1 Input

| | |
|---|---|
| **Supply** | MDB-1: transformer 1000 kVA, 5 % Z |
| **Loads** | Design loads × demand factors (load schedules and feeders) |

### 10.2 Design criteria

| | |
|---|---|
| **Voltage drop limit (source to load)** | 0.5 % |

### 10.3 Method

- Balanced 3-phase load flow from the design loads: demand = connected load × demand factor, summed up the network (kW and kVAr separately).
- Voltage drop per cable: $\sqrt{3} \cdot I \cdot L \cdot (R \cos\varphi + X \sin\varphi)$ for 3-phase, $2 \cdot I \cdot L \cdot (\ldots)$ for single-phase; busbar voltage = nominal minus the drops of the incomers above it. Limit 0.5 % from the main board to the load.
- Transformer regulation is not included (busbar voltages are measured from the main LV busbar).
- Motors: drop while starting = upstream drop + cable drop × the starter's starting-current multiple; limit 10 %.

### 10.4 Result

| Total demand of the scope | Lowest busbar voltage | Circuits checked |
|---|---|---|
| **451 kW · 512 kVA** | **99.08 %** | **0 pass · 1 warning · 14 fail** |

### 10.5 Busbars

**Table 14**

| Board | Connected (kW) | Demand (kW) | Demand (kVA) | PF | Current (A) | Rating (A) | Loading | Voltage (V) | % of nominal |
|---|---|---|---|---|---|---|---|---|---|
| MDB-1 | 514 | 451.1 | 512.1 | 0.88 | 712 | 1600 | 45 % | 415 | 100 |
| SMDB-GF | 231 | 203.3 | 233.7 | 0.87 | 325 | 630 | 52 % | 413.1 | 99.54 |
| DB-GF1 | 38 | 38 | 42.2 | 0.9 | 59 | 100 | 59 % | 411.2 | 99.08 |
| SMDB-FF | 109 | 87 | 97.4 | 0.89 | 135 | 250 | 54 % | 411.9 | 99.25 |
| MCC-1 | 130 | 130 | 151.2 | 0.86 | 210 | 400 | 53 % | 412.6 | 99.42 |

### 10.6 Motor starting — limit 10 % source to motor

**Table 15**

| Circuit | From | Motor | Running ΔV total (%) | Starting ΔV total (%) | Limit (%) | Result |
|---|---|---|---|---|---|---|
| MCC-WP | MCC-1 | Motor (DOL) | 1.03 | 3.3 | 10 | PASS |
| MCC-FP | MCC-1 | Fire pump (DOL) | 1.12 | 3.8 | 10 | PASS |

### 10.7 Verification — Voltage drop — worst circuit on each board

**Table 16**

| Board | Circuit | Calculated ΔV total (%) | Allowable (%) | Margin (%) | Status |
|---|---|---|---|---|---|
| MDB-1 | DB-EV | 2.38 | 0.5 | -1.88 | FAIL |
| SMDB-GF | GF-SKT | 2.37 | 0.5 | -1.87 | FAIL |
| SMDB-FF | FF-LTG | 2.57 | 0.5 | -2.07 | FAIL |
| MCC-1 | MCC-FP | 1.12 | 0.5 | -0.62 | FAIL |

*See also Appendix B (cable calculations).*

## 11. SHORT CIRCUIT STUDY

Scope: Whole installation (5 boards)

Overall status: **PASS** (36 pass · 0 warning · 0 fail)

### 11.1 Input

| | |
|---|---|
| **Supply** | MDB-1: transformer 1000 kVA, 5 % Z |
| **System** | 415 V, 50 Hz |

### 11.2 Design criteria

| | |
|---|---|
| **Voltage factor** | c max = 1.05 (maximum fault, cable R at 20 °C) |
| **Breaking capacity** | Breaker Icu ≥ prospective fault at its terminals (Ik″) |

### 11.3 Method

- Maximum prospective symmetrical 3-phase fault current Ik″, simplified method based on IEC 60909-0 (preliminary — not a full IEC 60909 study): voltage factor c max = 1.05, cable R at 20 °C, upstream MV network infinite, transformer correction factor KT not applied.
- Source impedance from the transformer rating and %Z (split by its X/R ratio); cable R at 20 °C and X per km; parallel runs divide the impedance.
- Breaking capacity check: the fault at the breaker terminals (supply busbar) must not exceed the breaker Icu. Fault at the cable end: 3-phase for 3/4-core circuits, line-to-neutral for 2-core.
- Motor contribution and generation are neglected.

### 11.4 Result

Highest busbar fault
**29.2 kA**

Breakers checked
**36 pass · 0 warning · 0 fail**

### 11.5 Fault levels at the busbars

**Table 17**

| Board | Type | Fed from | Busbar rating (A) | Ik″ max (kA) |
|---|---|---|---|---|
| MDB-1 | MDB | Transformer 1000 kVA, 5 % Z | 1600 | 29.2 |
| SMDB-GF | SMDB | MDB-1 | 630 | 21.8 |
| DB-GF1 | DB | SMDB-GF | 100 | 10.7 |
| SMDB-FF | SMDB | MDB-1 | 250 | 14.3 |
| MCC-1 | MCC | MDB-1 | 400 | 19 |

### 11.6 Verification — Breaking capacity at each busbar

**Table 18**

| Location | Ik″ busbar (kA) | Governing circuit | Ik″ at breaker (kA) | Icu (kA) | Margin (kA) | Breakers pass / check / fail | Status |
|---|---|---|---|---|---|---|---|
| MDB-1 | 29.2 | INC-GF | 29.2 | 36 | 6.8 | 5 / 0 / 0 | PASS |
| SMDB-GF | 21.8 | GF-LTG | 21.8 | 25 | 3.2 | 4 / 0 / 0 | PASS |
| DB-GF1 | 10.7 | DB-GF1-Y1 | 10.7 | 15 | 4.3 | 21 / 0 / 0 | PASS |
| SMDB-FF | 14.3 | FF-OFF | 14.3 | 25 | 10.7 | 4 / 0 / 0 | PASS |
| MCC-1 | 19 | MCC-WP | 19 | 25 | 6 | 2 / 0 / 0 | PASS |

*See also Appendix C (short-circuit and earth fault calculations).*

## 12. PROTECTION DISCRIMINATION STUDY

Scope: Whole installation (5 boards)

Overall status: **WARNING** (0 pass · 31 warning · 0 fail)

### 12.1 Design criteria

| | |
|---|---|
| **Overload selectivity** | In upstream ÷ In downstream ≥ 1.6 |
| **Short-circuit selectivity** | Fault at the downstream breaker below the upstream magnetic threshold |

### 12.2 Method

- Each breaker against the breaker above it. Overload (time) selectivity: In upstream ÷ In downstream ≥ 1.6.
- Short-circuit selectivity: total when the maximum fault at the downstream breaker stays below the upstream magnetic no-trip threshold; otherwise partial up to that current. Confirm with the manufacturer's selectivity tables.

### 12.3 Result

Pairs checked

**0 pass · 31 warning · 0 fail**

### 12.4 Selectivity

**Table 19**

| Upstream | Downstream | In ratio | Fault at downstream (kA) | Selectivity limit (kA) | Short circuit | Result |
|---|---|---|---|---|---|---|
| INC-GF (500 A) | GF-LTG (80 A) | 6.25 | 21.8 | 4 | Partial | **WARNING** |
| INC-GF (500 A) | GF-SKT (40 A) | 12.5 | 21.8 | 4 | Partial | **WARNING** |
| INC-GF (500 A) | GF-HVAC (250 A) | 2 | 21.8 | 4 | Partial | **WARNING** |
| INC-GF (500 A) | INC-DBGF1 (100 A) | 5 | 21.8 | 4 | Partial | **WARNING** |
| INC-FF (200 A) | FF-OFF (63 A) | 3.17 | 14.3 | 1.6 | Partial | **WARNING** |
| INC-FF (200 A) | FF-IT (40 A) | 5 | 14.3 | 1.6 | Partial | **WARNING** |
| INC-FF (200 A) | FF-LAB (80 A) | 2.5 | 14.3 | 1.6 | Partial | **WARNING** |
| INC-FF (200 A) | FF-LTG (25 A) | 8 | 14.3 | 1.6 | Partial | **WARNING** |
| INC-MCC (315 A) | MCC-WP (160 A) | 1.97 | 19 | 2.5 | Partial | **WARNING** |
| INC-MCC (315 A) | MCC-FP (125 A) | 2.52 | 19 | 2.5 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-Y1 (6 A) | 16.67 | 10.7 | 0.8 | Partial | **WARNING** |

| Upstream | Downstream | In ratio | Fault at downstream (kA) | Selectivity limit (kA) | Short circuit | Result |
|---|---|---|---|---|---|---|
| INC-DBGF1 (100 A) | DB-GF1-R1 (10 A) | 10 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-Y2 (6 A) | 16.67 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-B6 (10 A) | 10 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-Y5 (16 A) | 6.25 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-B7 (10 A) | 10 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-R3 (40 A) | 2.5 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-B3 (20 A) | 5 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-R6 (6 A) | 16.67 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-B4 (16 A) | 6.25 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-Y4 (16 A) | 6.25 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-B5 (16 A) | 6.25 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-R5 (10 A) | 10 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-Y8 (6 A) | 16.67 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-Y3 (20 A) | 5 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-R4 (16 A) | 6.25 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-B2 (6 A) | 16.67 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-Y7 (6 A) | 16.67 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-B1 (6 A) | 16.67 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-R7 (6 A) | 16.67 | 10.7 | 0.8 | Partial | **WARNING** |
| INC-DBGF1 (100 A) | DB-GF1-Y6 (16 A) | 6.25 | 10.7 | 0.8 | Partial | **WARNING** |

## 13. EARTH FAULT LOOP IMPEDANCE AND DISCONNECTION

Scope: Whole installation (5 boards)

Overall status: **PASS** (36 pass · 0 warning · 0 fail)

### 13.1 Input

| | |
|---|---|
| **Earthing arrangement** | TN-S |
| **Supply loop (Ze)** | From the source impedance to each board |

### 13.2 Design criteria

| | |
|---|---|
| **Disconnection time** | IEC 60364-4-41 Table 41.1: 0.4 s final circuits, 5 s distribution circuits |
| **Fault current** | Minimum, c = 0.95 |
| **Protective conductor** | Adiabatic, k = 143 |

### 13.3 Method

- TN-S system. Earth fault loop impedance Zs = Ze (at the supply board) + (R phase + R cpc) of the circuit, at operating temperature; minimum fault current with c = 0.95.
- Protective conductor (adiabatic): $S \geq I \cdot \sqrt{t} \div 143$, t = 0.1 s when tripping instantaneously (not the device let-through energy), with I the current through the conductor checked. Parallel runs: each CPC carries an equal share of the end-of-circuit fault (identical runs bonded at both ends); a result that passes only on that assumption is marked "assumed sharing" (warning) because a fault within one run or unequal runs are not covered. The total fault current sets the disconnection check.
- Disconnection per IEC 60364-4-41 Table 41.1 (TN, U0 in the 230 V band): 0.4 s for final circuits up to 63 A with socket-outlets and up to 32 A supplying fixed equipment only; 5 s for distribution circuits and other final circuits. Circuits whose purpose is not set are treated as socket-outlet circuits. Instantaneous tripping when If ≥ Ia; below that, a 5 s circuit needs the breaker's time-current curve checked. Protective conductor checked by the adiabatic equation (k = 143).

### 13.4 Result

Circuits checked

**36 pass · 0 warning · 0 fail**

### 13.5 Verification — Earth fault loop on each board

**Table 20**

| Board | Circuits | Highest Zs ÷ max Zs | Pass / check / fail | Status |
|---|---|---|---|---|
| MDB-1 | 5 | 74 % | 5 / 0 / 0 | PASS |
| SMDB-GF | 4 | 70 % | 4 / 0 / 0 | PASS |
| DB-GF1 | 21 | 0 % | 21 / 0 / 0 | PASS |

| Board | Circuits | Highest Zs ÷ max Zs | Pass / check / fail | Status |
|---|---|---|---|---|
| SMDB-FF | 4 | 46 % | 4 / 0 / 0 | PASS |
| MCC-1 | 2 | 38 % | 2 / 0 / 0 | PASS |

*See also Appendix C (short-circuit and earth fault calculations).*

## 14. PHASE BALANCE

Scope: Whole installation (5 boards)

Overall status: **WARNING** (3 pass · 2 warning · 0 fail)

### 14.1 Method

- Demand kW and kvar per phase (each load's demand factor and power factor), downstream boards included: single-phase circuits on their labelled phase R / Y / B; 3-phase loads spread evenly.
- Phase current I = S ÷ U0 at nominal voltage (U0 = 240 V). Neutral current = phasor sum of the three phase currents, phases 120° apart, at fundamental frequency — triplen harmonics (e.g. from LED drivers and IT loads) add to the neutral and are not included.
- Current unbalance = largest deviation from the average phase current ÷ the average. Flagged above 10 % (app default, not a standard's limit).
- Limits of this check: demand-based at nominal voltage — not an unbalanced load flow; no voltage unbalance, no neutral voltage, no phase-specific voltage drop. Single-phase circuits with no phase set are spread evenly, which hides their unbalance; they are listed.

### 14.2 Result

| Boards checked | Single-phase circuits without a phase |
|---|---|
| **3 pass · 2 warning · 0 fail** | **1 (FF-LTG) — spread evenly, unbalance may be understated** |

### 14.3 Phase balance at each board (demand)

**Table 21**

| Board | R (A) | Y (A) | B (A) | R / Y / B (kW) | Neutral (A) | Unbalance | No phase set | Result |
|---|---|---|---|---|---|---|---|---|
| MDB-1 | 776.1 | 774.2 | 773 | 167.4 / 167 / 166.7 | 2.7 | 0.2 % | **1** | **WARNING** |
| SMDB-GF | 326.8 | 324.9 | 323.7 | 68.1 / 67.7 / 67.5 | 2.7 | 0.5 % | 0 | PASS |
| DB-GF1 | 60.4 | 58.4 | 57.3 | 13 / 12.6 / 12.3 | 2.7 | 2.9 % | 0 | PASS |
| SMDB-FF | 135.5 | 135.5 | 135.5 | 29 / 29 / 29 | 0 | 0 % | **1** | **WARNING** |
| MCC-1 | 210.3 | 210.3 | 210.3 | 43.3 / 43.3 / 43.3 | 0 | 0 % | 0 | PASS |

## 15. BUSBAR TRUNKING RISERS

Scope: Whole installation (5 boards)

### 15.1 Method

- Demand: the tap-offs' loads (a floor's board demand, or the load entered) × diversity (IEC 61439-6: 0.9 for 2–3 tap-offs, 0.8 for 4–5, 0.7 for 6–9, 0.6 for 10 or more, unless set).
- Rating: Ib ≤ In (feeding breaker) ≤ busway rating × ambient derating (≈ 1 % per °C above 40 °C).
- Voltage drop floor by floor: $\Delta U = \sqrt{3} \cdot I \cdot L \cdot (R \cos \varphi + X \sin \varphi)$, each riser section carrying the diversified load of the tap-offs above it.
- Short-circuit: the busway's Icw (1 s) ≥ the fault level at the board feeding it. Busway data: typical sandwich busway — confirm with the manufacturer.

### 15.2 Result

Risers checked

**0 pass · 0 warning · 0 fail**

### 15.3 Busbar risers

Nothing in scope.

## 16. POWER FACTOR CORRECTION

Scope: Whole installation (5 boards)

Overall status: **WARNING** (0 pass · 1 warning · 0 fail)

### 16.1 Method

- Strategy: Central — at the main board (MDB).
- Qc = P × (tan φ1 − tan φ2) to PF 0.95, from the maximum demand (demand factors applied). Capacitors already on the drawing are included.
- Banks are sized from the bottom up: each bank only covers what the banks below it do not, so nothing is corrected twice.
- Board banks: automatic (APFC) in 25 kvar steps, rounded up. Individual capacitors: fixed, rounded down (never over-correct a motor).
- Detuned (7 %, 189 Hz) where non-linear load ≥ 25 % of demand. Capacitor rating ≥ 1.05 × U ÷ (1 − p).
- Breaker ≥ 1.43 × bank current (IEC 60831 1.3 × overcurrent and capacitance tolerance). Light-load check at 30 % of demand.

### 16.2 Result

Target PF
**0.95**

Capacitors (new)
**100 kvar in 1 banks**

MDB-1 PF
**0.88 → 0.954**

MDB-1 capacity released
**39 kVA (transformer 51 % → 47 %)**

### 16.3 Capacitor banks

**Table 22**

| Location | Demand (kW) | Reactive (kvar) | Existing (kvar) | Banks below (kvar) | PF now | Required (kvar) | Bank | Detuning · capacitor V | Breaker · cable | PF after | Current (A) | Result |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| MDB-1 | 451 | 243 | — | — | 0.88 | 94.2 | 100 kvar (4 × 25) | None · 440 V | 200 A · 4C × 70 mm² | 0.954 | 712 → 658 | **WARNING** |

### 16.4 Main boards

**Table 23**

| Main board | Demand (kW) | PF before | PF after | kVA before → after | Released (kVA) | Transformer loading |
|---|---|---|---|---|---|---|
| MDB-1 | 451 | 0.88 | 0.954 | 512 → 473 | 39 | 51 % → 47 % |

### 16.5 Notes

**Table 24**

| Location | Note |
|---|---|

| Location | Note |
|---|---|
| MDB-1 | 17 % non-linear load — check harmonics; consider a detuned bank |

## 17. DB AND CABLE SCHEDULES

Scope: Whole installation (5 boards)

### 17.1 Method

- Schedules of the boards in scope, from the design data and the latest calculation run.

### 17.2 Result

| Boards |
| --- |
| **5** |

*See also Appendix A (detailed load schedule), Appendix B (cable calculations).*

## 18. RESULTS SUMMARY

### 18.1 Transformers

**Table 25**

| Main board | Installed (kVA) | %Z | Required (kVA) | Demand (kVA) | Loading | Status |
|---|---|---|---|---|---|---|
| MDB-1 | 1000 | 5 | 710 | 473 | 47 % | PASS |

### 18.2 Standby generator

**Table 26**

| Essential demand (kVA) | Required (kVA) | Recommended | Installed (kVA) | Status |
|---|---|---|---|---|
| 64 | 544 | 630 kVA | Not defined | *DATA REQUIRED* |

### 18.3 Distribution boards (MDB, SMDB, DB, MCC)

**Table 27**

| Board | Type | Bus rating (A) | Demand (A) | Loading | Ik″ (kA) | Status |
|---|---|---|---|---|---|---|
| MDB-1 | MDB | 1600 | 712 | 45 % | 29.2 | PASS |
| SMDB-GF | SMDB | 630 | 325 | 52 % | 21.8 | PASS |
| DB-GF1 | DB | 100 | 59 | 59 % | 10.7 | PASS |
| SMDB-FF | SMDB | 250 | 135 | 54 % | 14.3 | PASS |
| MCC-1 | MCC | 400 | 210 | 53 % | 19 | PASS |

### 18.4 Capacitor banks

**Table 28**

| Location | Bank (kvar) | Steps | PF before → after | Target PF | Status |
|---|---|---|---|---|---|
| MDB-1 | 100 | 4 × 25 | 0.88 → 0.954 | 0.95 | **WARNING** |

### 18.5 Load summary

**Table 29**

| Item | Value |
|---|---|
| Connected load | 514 kW |
| Maximum demand | 451 kW · 512 kVA |
| Transformer MDB-1 loading | 47 % of 1000 kVA (design limit 80 %) |

| Item | Value |
|---|---|
| Transformer MDB-1 spare | 527 kVA |
| Generator loading | Installed rating not defined |

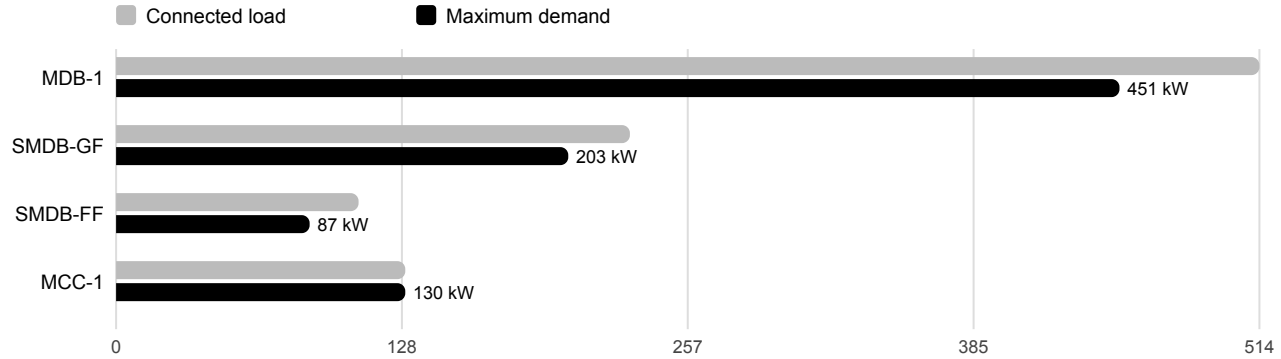



*Figure 1 — Connected load vs maximum demand. Per board, kW. The gap is the diversity applied by the demand factors.*




*Figure 2 — Transformer capacity vs demand. Demand against the installed rating (tick) and the 80 % design loading limit (dashed).*

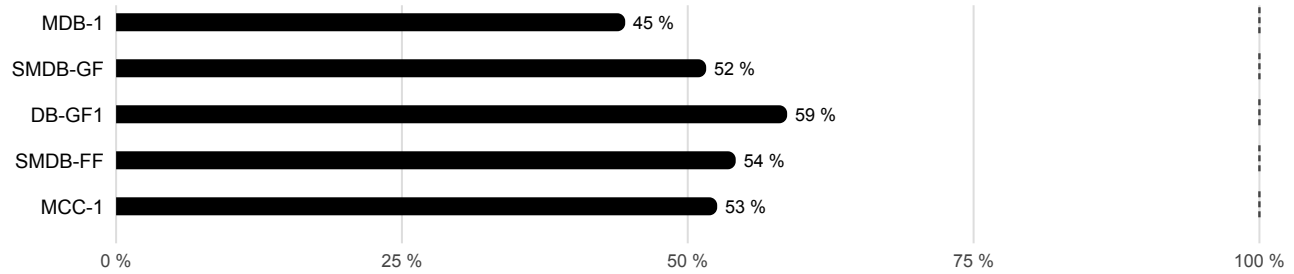



*Figure 3 — Board loading. Demand current ÷ bus rating. Boards without a rating are not shown.*

## 18.6 Cable summary

**Table 30**

| From | To | Load (kW) | Current (A) | Cable | Protective device | Voltage drop (%) | Status |
|---|---|---|---|---|---|---|---|
| MDB-1 | SMDB-GF | — | 327 | 4C × 300 mm² | 500 A MCCB | 0.46 | **WARNING** |
| MDB-1 | SMDB-FF | — | 135 | 4C × 95 mm² | 200 A MCCB | 0.75 | **FAIL** |

| From | To | Load (kW) | Current (A) | Cable | Protective device | Voltage drop (%) | Status |
|---|---|---|---|---|---|---|---|
| MDB-1 | MCC-1 | — | 210 | 4C × 150 mm² | 315 A MCCB | 0.58 | FAIL |
| MDB-1 | EV charging – parking | 30.8 | 44 | 4C × 10 mm² | 63 A C | 2.38 | FAIL |
| MDB-1 | Solar PV inverter | 50 | 70 | 4C × 25 mm² | 100 A MCCB | 0.76 | FAIL |
| SMDB-GF | Lighting | 40.5 | 63 | 4C × 16 mm² | 80 A MCCB | 1.47 | FAIL |
| SMDB-GF | Sockets | 16.8 | 27 | 4C × 4 mm² | 40 A C | 2.37 | FAIL |
| SMDB-GF | HVAC | 108 | 177 | 4C × 95 mm² | 250 A MCCB | 1.16 | FAIL |
| SMDB-FF | Office | 28 | 43 | 4C × 10 mm² | 63 A C | 1.66 | FAIL |
| SMDB-FF | IT room | 20 | 29 | 4C × 4 mm² | 40 A C | 2.04 | FAIL |
| SMDB-FF | Lab | 35 | 57 | 4C × 16 mm² | 80 A MCCB | 1.62 | FAIL |
| SMDB-FF | Corridor lighting | 4 | 19 | 2C × 6 mm² | 25 A C | 2.57 | FAIL |
| MCC-1 | Water pump | 75 | 121 | 4C × 50 mm² | 160 A MCCB | 1.03 | FAIL |
| MCC-1 | Fire pump | 55 | 89 | 4C × 35 mm² | 125 A MCCB | 1.12 | FAIL |
| SMDB-GF | DB-GF1 | — | 60 | 4C × 35 mm² | 100 A MCCB | 0.92 | FAIL |

*Final circuits are on the DB load schedules.*

## 19. COMPLIANCE SUMMARY

Overall design status: **FAIL** (7 PASS · 1 WARNING · 1 FAIL · 1 DATA REQUIRED)

**Table 31**

| Design check | Governing item | Calculated | Requirement / selected | Margin | Status |
|---|---|---|---|---|---|
| Transformer loading | MDB-1 | 473 kVA | 1000 kVA | 527 kVA | PASS |
| Generator loading | Standby generator | 64 kVA | Installed rating not defined | — | *DATA REQUIRED* |
| Fault level (breaking capacity) | GF-LTG | 21.8 kA | 25 kA | 3.2 kA | PASS |
| Voltage drop | DB-GF1-R3 | 2.79 % | 0.5 % | -2.29 % | **FAIL** |
| Cable capacity (Ib ≤ Iz) | FF-IT | 29 A | 41 A | 12 A | PASS |
| Overload protection (Ib ≤ In ≤ Iz) | 36 circuits | 36 of 36 meet it | All circuits | — | PASS |
| Board loading | DB-GF1 | 59 A | 100 A | 41 A | PASS |
| Earth fault disconnection | DB-EV | Zs 0.266 Ω | ≤ 0.361 Ω | 0.096 Ω | PASS |
| Selectivity | 31 breaker pairs | 0 total, 31 partial / to check | Total selectivity | — | **WARNING** |
| Power factor | MDB-1 | 0.954 | ≥ 0.95 | 0.004 | PASS |

*Each row shows the item with the least margin; the status is the worst over all items of that check. PASS / WARNING / FAIL are the app's calculated checks against the project limits; NOT CHECKED: nothing installed or selected to check against; DATA REQUIRED: a rating or input is missing. Engineering review and approval are separate.*

# APPENDIX A — DETAILED LOAD SCHEDULE

## A.1 DB and cable schedules — DB schedule

**Table 32**

| Board | Cct | Circuit ID | Description | Load type | Connected (kW) | DF | Demand (kW) | PF | Ib (A) | Breaker | Poles | Icu (kA) | Cable | Length (m) | Vd total (%) | Status | Accessories |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| MDB-1 | 1 | INC-GF | Incomer to Ground floor SMDB | Sub-board SMDB-GF | | | | | 327 | 500 A MCCB | TP+N | 36 | 4C × 300 mm² XLPE/SWA + 150 mm² CPC | 35 | 0.46 | **WARNING** | |
| MDB-1 | 2 | INC-FF | Incomer to First floor SMDB | Sub-board SMDB-FF | | | | | 135 | 200 A MCCB | TP+N | 36 | 4C × 95 mm² XLPE/SWA + 50 mm² CPC | 55 | 0.75 | **FAIL** | |
| MDB-1 | 3 | INC-MCC | Incomer to Pump room MCC | Sub-board MCC-1 | | | | | 210 | 315 A MCCB | TP+N | 36 | 4C × 150 mm² XLPE/SWA + 95 mm² CPC | 40 | 0.58 | **FAIL** | |
| MDB-1 | 4 | DB-EV | EV charging – parking | EV charging | 44 | 0.7 | 30.8 | 0.98 | 44 | 63 A C | TP+N | 36 | 4C × 10 mm² XLPE/SWA + 10 mm² CPC | 60 | 2.38 | **FAIL** | |
| MDB-1 | 5 | DB-PV | Solar PV inverter | Solar PV | 50 | 1 | 50 | 1 | 70 | 100 A MCCB | TP+N | 36 | 4C × 25 mm² XLPE/SWA + 16 mm² CPC | 30 | 0.76 | **FAIL** | |
| MDB-1 | | TOTAL | Main Distribution Board — 1600 A | | 514 | | 451.1 | 0.88 | 712 | | | 29.2 | | | | 45% loaded | |
| SMDB-GF | 1 | GF-LTG | Lighting | Lighting | 45 | 0.9 | 40.5 | 0.9 | 63 | 80 A MCCB | TP+N | 25 | 4C × 16 mm² XLPE/SWA + 16 mm² CPC | 30 | 1.47 | **FAIL** | |
| SMDB-GF | 2 | GF-SKT | Sockets | Small power / sockets | 28 | 0.6 | 16.8 | 0.85 | 27 | 40 A C | TP+N | 25 | 4C × 4 mm² XLPE/SWA + 4 mm² CPC | 35 | 2.37 | **FAIL** | |
| SMDB-GF | 3 | GF-HVAC | HVAC | HVAC | 120 | 0.9 | 108 | 0.85 | 177 | 250 A MCCB | TP+N | 25 | 4C × 95 mm² XLPE/SWA + 50 mm² CPC | 40 | 1.16 | **FAIL** | |
| SMDB-GF | 4 | INC-DBGF1 | Incomer to Villa ground floor DB | Sub-board DB-GF1 | | | | | 60 | 100 A MCCB | TP+N | 25 | 4C × 35 mm² XLPE/SWA + 16 mm² CPC | 30 | 0.92 | **FAIL** | |

| Board | Cct | Circuit ID | Description | Load type | Connected (kW) | DF | Demand (kW) | PF | Ib (A) | Breaker | Poles | Icu (kA) | Cable | Length (m) | Vd total (%) | Status | Accessories |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SMDB-GF | | TOTAL | Ground floor SMDB — 630 A | | 231 | | 203.3 | 0.87 | 325 | | | 21.8 | | | | 52% loaded | |
| DB-GF1 | 1 | DB-GF1-Y1 | Majlis lighting | Lighting | 1 | 1 | 1 | 0.9 | 5 | 6 A C | SP+N | 15 | 2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC | 18 | 1.48 | FAIL | |
| DB-GF1 | 2 | DB-GF1-R1 | Living lighting | Lighting | 1.2 | 1 | 1.2 | 0.9 | 6 | 10 A C | SP+N | 15 | 2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC | 20 | 1.67 | FAIL | |
| DB-GF1 | 3 | DB-GF1-Y2 | Kitchen lighting | Lighting | 0.8 | 1 | 0.8 | 0.9 | 4 | 6 A C | SP+N | 15 | 2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC | 22 | 1.49 | FAIL | |
| DB-GF1 | 4 | DB-GF1-B6 | Majlis sockets | Small power / sockets | 1.5 | 1 | 1.5 | 0.9 | 7 | 10 A C | SP+N | 15 | 2C × 4 mm² XLPE/SWA + 4 mm² CPC | 18 | 1.44 | FAIL | |
| DB-GF1 | 5 | DB-GF1-Y5 | Living sockets | Small power / sockets | 2 | 1 | 2 | 0.9 | 9 | 16 A C | SP+N | 15 | 2C × 4 mm² XLPE/SWA + 4 mm² CPC | 20 | 1.7 | FAIL | |
| DB-GF1 | 6 | DB-GF1-B7 | Kitchen sockets | Small power / sockets | 1.3 | 1 | 1.3 | 0.9 | 6 | 10 A C | SP+N | 15 | 2C × 4 mm² XLPE/SWA + 4 mm² CPC | 22 | 1.45 | FAIL | |
| DB-GF1 | 7 | DB-GF1-R3 | Cooker | General | 6 | 1 | 6 | 0.9 | 28 | 40 A C | SP+N | 15 | 2C × 6 mm² XLPE/SWA + 6 mm² CPC | 24 | 2.79 | FAIL | |
| DB-GF1 | 8 | DB-GF1-B3 | Kitchen water heater | General | 3 | 1 | 3 | 0.9 | 14 | 20 A C | SP+N | 15 | 2C × 4 mm² XLPE/SWA + 4 mm² CPC | 24 | 2.32 | FAIL | |
| DB-GF1 | 9 | DB-GF1-R6 | Maid room | Small power / sockets | 1.1 | 1 | 1.1 | 0.9 | 5 | 6 A C | SP+N | 15 | 2C × 4 mm² XLPE/SWA + 4 mm² CPC | 28 | 1.51 | FAIL | |
| DB-GF1 | 10 | DB-GF1-B4 | Majlis split A/C | HVAC | 2.5 | 1 | 2.5 | 0.9 | 12 | 16 A C | SP+N | 15 | 2C × 4 mm² XLPE/SWA + 4 mm² CPC | 18 | 1.79 | FAIL | |

| Board | Cct | Circuit ID | Description | Load type | Connected (kW) | DF | Demand (kW) | PF | Ib (A) | Breaker | Poles | Icu (kA) | Cable | Length (m) | Vd total (%) | Status | Accessories |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DB-GF1 | 11 | DB-GF1-Y4 | Living split A/C | HVAC | 2.5 | 1 | 2.5 | 0.9 | 12 | 16 A C | SP+N | 15 | 2C × 4 mm² XLPE/SWA + 4 mm² CPC | 20 | 1.89 | FAIL | |
| DB-GF1 | 12 | DB-GF1-B5 | Dining split A/C | HVAC | 2.5 | 1 | 2.5 | 0.9 | 12 | 16 A C | SP+N | 15 | 2C × 4 mm² XLPE/SWA + 4 mm² CPC | 22 | 1.99 | FAIL | |
| DB-GF1 | 13 | DB-GF1-R5 | Guest bedroom | Small power / sockets | 1.5 | 1 | 1.5 | 0.9 | 7 | 10 A C | SP+N | 15 | 2C × 4 mm² XLPE/SWA + 4 mm² CPC | 26 | 1.67 | FAIL | |
| DB-GF1 | 14 | DB-GF1-Y8 | Guest bathroom | Lighting | 0.3 | 1 | 0.3 | 0.9 | 1 | 6 A C | SP+N | 15 | 2C × 4 mm² XLPE/SWA + 4 mm² CPC | 27 | 1.05 | FAIL | |
| DB-GF1 | 15 | DB-GF1-Y3 | Guest bath water heater | General | 3 | 1 | 3 | 0.9 | 14 | 20 A C | SP+N | 15 | 2C × 4 mm² XLPE/SWA + 4 mm² CPC | 27 | 2.49 | FAIL | |
| DB-GF1 | 16 | DB-GF1-R4 | Guest bedroom split A/C | HVAC | 2.5 | 1 | 2.5 | 0.9 | 12 | 16 A C | SP+N | 15 | 2C × 4 mm² XLPE/SWA + 4 mm² CPC | 26 | 2.18 | FAIL | |
| DB-GF1 | 17 | DB-GF1-B2 | Dining lighting | Lighting | 0.6 | 1 | 0.6 | 0.9 | 3 | 6 A C | SP+N | 15 | 2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC | 22 | 1.33 | FAIL | |
| DB-GF1 | 18 | DB-GF1-Y7 | Dining sockets | Small power / sockets | 1 | 1 | 1 | 0.9 | 5 | 6 A C | SP+N | 15 | 2C × 4 mm² XLPE/SWA + 4 mm² CPC | 22 | 1.35 | FAIL | |
| DB-GF1 | 19 | DB-GF1-B1 | External lighting | Lighting | 1 | 1 | 1 | 0.9 | 5 | 6 A C | SP+N | 15 | 2C × 2.5 mm² XLPE/SWA + 2.5 mm² CPC | 35 | 2.01 | FAIL | |
| DB-GF1 | 20 | DB-GF1-R7 | Water pump | Motor | 0.8 | 1 | 0.8 | 0.9 | 3 | 6 A C | SP+N | 15 | 2C × 4 mm² XLPE/SWA + 4 mm² CPC | 30 | 1.36 | FAIL | |
| DB-GF1 | 21 | DB-GF1-Y6 | Laundry | Small power / sockets | 2 | 1 | 2 | 0.9 | 9 | 16 A C | SP+N | 15 | 2C × 4 mm² XLPE/SWA + 4 mm² CPC | 25 | 1.89 | FAIL | |

| Board | Cct | Circuit ID | Description | Load type | Connected (kW) | DF | Demand (kW) | PF | Ib (A) | Breaker | Poles | Icu (kA) | Cable | Length (m) | Vd total (%) | Status | Accessories |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DB-GF1 | | TOTAL | Villa ground floor DB — 100 A | | 38 | | 38 | 0.9 | 59 | | | 10.7 | | | | 59% loaded | |
| SMDB-FF | 1 | FF-OFF | Office | Small power / sockets | 35 | 0.8 | 28 | 0.9 | 43 | 63 A C | TP+N | 25 | 4C × 10 mm² XLPE/SWA + 10 mm² CPC | 25 | 1.66 | FAIL | |
| SMDB-FF | 2 | FF-IT | IT room | IT / data | 20 | 1 | 20 | 0.95 | 29 | 40 A C | TP+N | 25 | 4C × 4 mm² XLPE/SWA + 4 mm² CPC | 20 | 2.04 | FAIL | |
| SMDB-FF | 3 | FF-LAB | Lab | General | 50 | 0.7 | 35 | 0.85 | 57 | 80 A MCCB | TP+N | 25 | 4C × 16 mm² XLPE/SWA + 16 mm² CPC | 30 | 1.62 | FAIL | |
| SMDB-FF | 4 | FF-LTG | Corridor lighting | Lighting | 4 | 1 | 4 | 0.9 | 19 | 25 A C | SP+N | 25 | 2C × 6 mm² XLPE/SWA + 6 mm² CPC | 35 | 2.57 | FAIL | |
| SMDB-FF | | TOTAL | First floor SMDB — 250 A | | 109 | | 87 | 0.89 | 135 | | | 14.3 | | | | 54% loaded | |
| MCC-1 | 1 | MCC-WP | Water pump | Motor | 75 | 1 | 75 | 0.86 | 121 | 160 A MCCB | TP+N | 25 | 4C × 50 mm² XLPE/SWA + 25 mm² CPC | 20 | 1.03 | FAIL | |
| MCC-1 | 2 | MCC-FP | Fire pump | Fire pump | 55 | 1 | 55 | 0.86 | 89 | 125 A MCCB | TP+N | 25 | 4C × 35 mm² FR + 16 mm² CPC | 25 | 1.12 | FAIL | |
| MCC-1 | | TOTAL | Pump room MCC — 400 A | | 130 | | 130 | 0.86 | 210 | | | 19 | | | | 53% loaded | |

# APPENDIX B — CABLE CALCULATIONS

## B.1 Cable and breaker sizing — Cables and breakers

**Table 33**

| Circuit | From | To | Cable | Installation | Derating | Ib (A) | Breaker In (A) | Iz (A) | Ib ≤ In ≤ Iz | ΔV total (%) | Result |
|---|---|---|---|---|---|---|---|---|---|---|---|
| INC-GF | MDB-1 | SMDB-GF | 4C × 300 mm² XLPE/SWA + 150 CPC | E | 0.87 · 1.00 | 326.8 | 500 MCCB | 525 | PASS | 0.46 | **WARNING** |
| INC-FF | MDB-1 | SMDB-FF | 4C × 95 mm² XLPE/SWA + 50 CPC | E | 0.87 · 1.00 | 135.5 | 200 MCCB | 266 | PASS | 0.75 | **FAIL** |
| INC-MCC | MDB-1 | MCC-1 | 4C × 150 mm² XLPE/SWA + 95 CPC | E | 0.87 · 1.00 | 210.3 | 315 MCCB | 342 | PASS | 0.58 | **FAIL** |
| DB-EV | MDB-1 | EV charging – parking | 4C × 10 mm² XLPE/SWA + 10 CPC | E | 0.87 · 1.00 | 43.7 | 63 C | 70 | PASS | 2.38 | **FAIL** |
| DB-PV | MDB-1 | Solar PV inverter | 4C × 25 mm² XLPE/SWA + 16 CPC | E | 0.87 · 1.00 | 69.6 | 100 MCCB | 120 | PASS | 0.76 | **FAIL** |
| GF-LTG | SMDB-GF | Lighting | 4C × 16 mm² XLPE/SWA + 16 CPC | E | 0.87 · 1.00 | 62.6 | 80 MCCB | 91 | PASS | 1.47 | **FAIL** |
| GF-SKT | SMDB-GF | Sockets | 4C × 4 mm² XLPE/SWA + 4 CPC | E | 0.87 · 1.00 | 27.5 | 40 C | 41 | PASS | 2.37 | **FAIL** |
| GF-HVAC | SMDB-GF | HVAC | 4C × 95 mm² XLPE/SWA + 50 CPC | E | 0.87 · 1.00 | 176.8 | 250 MCCB | 266 | PASS | 1.16 | **FAIL** |
| FF-OFF | SMDB-FF | Office | 4C × 10 mm² XLPE/SWA + 10 CPC | E | 0.87 · 1.00 | 43.3 | 63 C | 70 | PASS | 1.66 | **FAIL** |
| FF-IT | SMDB-FF | IT room | 4C × 4 mm² XLPE/SWA + 4 CPC | E | 0.87 · 1.00 | 29.3 | 40 C | 41 | PASS | 2.04 | **FAIL** |
| FF-LAB | SMDB-FF | Lab | 4C × 16 mm² XLPE/SWA + 16 CPC | E | 0.87 · 1.00 | 57.3 | 80 MCCB | 91 | PASS | 1.62 | **FAIL** |
| FF-LTG | SMDB-FF | Corridor lighting | 2C × 6 mm² XLPE/SWA + 6 CPC | E | 0.87 · 1.00 | 18.5 | 25 C | 52 | PASS | 2.57 | **FAIL** |
| MCC-WP | MCC-1 | Water pump | 4C × 50 mm² XLPE/SWA + 25 CPC | E | 0.87 · 1.00 | 121.3 | 160 MCCB | 174 | PASS | 1.03 | **FAIL** |
| MCC-FP | MCC-1 | Fire pump | 4C × 35 mm² FR + 16 CPC | E | 0.87 · 1.00 | 89 | 125 MCCB | 146 | PASS | 1.12 | **FAIL** |
| INC-DBGF1 | SMDB-GF | DB-GF1 | 4C × 35 mm² XLPE/SWA + 16 CPC | E | 0.87 · 1.00 | 60.4 | 100 MCCB | 146 | PASS | 0.92 | **FAIL** |
| DB-GF1-Y1 | DB-GF1 | Majlis lighting | 2C × 2.5 mm² XLPE/SWA + 2.5 CPC | E | 0.87 · 1.00 | 4.6 | 6 C | 31 | PASS | 1.48 | **FAIL** |
| DB-GF1-R1 | DB-GF1 | Living lighting | 2C × 2.5 mm² XLPE/SWA + 2.5 CPC | E | 0.87 · 1.00 | 5.6 | 10 C | 31 | PASS | 1.67 | **FAIL** |
| DB-GF1-Y2 | DB-GF1 | Kitchen lighting | 2C × 2.5 mm² XLPE/SWA + 2.5 CPC | E | 0.87 · 1.00 | 3.9 | 6 C | 31 | PASS | 1.49 | **FAIL** |
| DB-GF1-B6 | DB-GF1 | Majlis sockets | 2C × 4 mm² XLPE/SWA + 4 CPC | E | 0.87 · 1.00 | 7 | 10 C | 41 | PASS | 1.44 | **FAIL** |
| DB-GF1-Y5 | DB-GF1 | Living sockets | 2C × 4 mm² XLPE/SWA + 4 CPC | E | 0.87 · 1.00 | 9.3 | 16 C | 41 | PASS | 1.7 | **FAIL** |
| DB-GF1-B7 | DB-GF1 | Kitchen sockets | 2C × 4 mm² XLPE/SWA + 4 CPC | E | 0.87 · 1.00 | 5.8 | 10 C | 41 | PASS | 1.45 | **FAIL** |
| DB-GF1-R3 | DB-GF1 | Cooker | 2C × 6 mm² XLPE/SWA + 6 CPC | E | 0.87 · 1.00 | 27.8 | 40 C | 52 | PASS | 2.79 | **FAIL** |
| DB-GF1-B3 | DB-GF1 | Kitchen water heater | 2C × 4 mm² XLPE/SWA + 4 CPC | E | 0.87 · 1.00 | 13.9 | 20 C | 41 | PASS | 2.32 | **FAIL** |

| Circuit | From | To | Cable | Installation | Derating | Ib (A) | Breaker In (A) | Iz (A) | Ib ≤ In ≤ Iz | ΔV total (%) | Result |
|---|---|---|---|---|---|---|---|---|---|---|---|
| DB-GF1-R6 | DB-GF1 | Maid room | 2C × 4 mm² XLPE/SWA + 4 CPC | E | 0.87 · 1.00 | 5.1 | 6 C | 41 | PASS | 1.51 | FAIL |
| DB-GF1-B4 | DB-GF1 | Majlis split A/C | 2C × 4 mm² XLPE/SWA + 4 CPC | E | 0.87 · 1.00 | 11.6 | 16 C | 41 | PASS | 1.79 | FAIL |
| DB-GF1-Y4 | DB-GF1 | Living split A/C | 2C × 4 mm² XLPE/SWA + 4 CPC | E | 0.87 · 1.00 | 11.6 | 16 C | 41 | PASS | 1.89 | FAIL |
| DB-GF1-B5 | DB-GF1 | Dining split A/C | 2C × 4 mm² XLPE/SWA + 4 CPC | E | 0.87 · 1.00 | 11.6 | 16 C | 41 | PASS | 1.99 | FAIL |
| DB-GF1-R5 | DB-GF1 | Guest bedroom | 2C × 4 mm² XLPE/SWA + 4 CPC | E | 0.87 · 1.00 | 6.9 | 10 C | 41 | PASS | 1.67 | FAIL |
| DB-GF1-Y8 | DB-GF1 | Guest bathroom | 2C × 4 mm² XLPE/SWA + 4 CPC | E | 0.87 · 1.00 | 1.2 | 6 C | 41 | PASS | 1.05 | FAIL |
| DB-GF1-Y3 | DB-GF1 | Guest bath water heater | 2C × 4 mm² XLPE/SWA + 4 CPC | E | 0.87 · 1.00 | 13.9 | 20 C | 41 | PASS | 2.49 | FAIL |
| DB-GF1-R4 | DB-GF1 | Guest bedroom split A/C | 2C × 4 mm² XLPE/SWA + 4 CPC | E | 0.87 · 1.00 | 11.6 | 16 C | 41 | PASS | 2.18 | FAIL |
| DB-GF1-B2 | DB-GF1 | Dining lighting | 2C × 2.5 mm² XLPE/SWA + 2.5 CPC | E | 0.87 · 1.00 | 2.8 | 6 C | 31 | PASS | 1.33 | FAIL |
| DB-GF1-Y7 | DB-GF1 | Dining sockets | 2C × 4 mm² XLPE/SWA + 4 CPC | E | 0.87 · 1.00 | 4.6 | 6 C | 41 | PASS | 1.35 | FAIL |
| DB-GF1-B1 | DB-GF1 | External lighting | 2C × 2.5 mm² XLPE/SWA + 2.5 CPC | E | 0.87 · 1.00 | 4.6 | 6 C | 31 | PASS | 2.01 | FAIL |
| DB-GF1-R7 | DB-GF1 | Water pump | 2C × 4 mm² XLPE/SWA + 4 CPC | E | 0.87 · 1.00 | 3.5 | 6 C | 41 | PASS | 1.36 | FAIL |
| DB-GF1-Y6 | DB-GF1 | Laundry | 2C × 4 mm² XLPE/SWA + 4 CPC | E | 0.87 · 1.00 | 9.3 | 16 C | 41 | PASS | 1.89 | FAIL |

## B.2 Load flow and voltage drop study — Feeders

**Table 34**

| Circuit | From | To | Cable | Length (m) | Ib (A) | PF | ΔV cable (%) | ΔV total (%) | Result |
|---|---|---|---|---|---|---|---|---|---|
| INC-GF | MDB-1 | SMDB-GF | 4C × 300 mm² | 35 | 326.8 | 0.87 | 0.46 | 0.46 | WARNING |
| INC-FF | MDB-1 | SMDB-FF | 4C × 95 mm² | 55 | 135.5 | 0.89 | 0.75 | 0.75 | FAIL |
| INC-MCC | MDB-1 | MCC-1 | 4C × 150 mm² | 40 | 210.3 | 0.86 | 0.58 | 0.58 | FAIL |
| DB-EV | MDB-1 | EV charging – parking | 4C × 10 mm² | 60 | 43.7 | 0.98 | 2.38 | 2.38 | FAIL |
| DB-PV | MDB-1 | Solar PV inverter | 4C × 25 mm² | 30 | 69.6 | 1 | 0.76 | 0.76 | FAIL |
| GF-LTG | SMDB-GF | Lighting | 4C × 16 mm² | 30 | 62.6 | 0.9 | 1 | 1.47 | FAIL |
| GF-SKT | SMDB-GF | Sockets | 4C × 4 mm² | 35 | 27.5 | 0.85 | 1.91 | 2.37 | FAIL |
| GF-HVAC | SMDB-GF | HVAC | 4C × 95 mm² | 40 | 176.8 | 0.85 | 0.7 | 1.16 | FAIL |
| FF-OFF | SMDB-FF | Office | 4C × 10 mm² | 25 | 43.3 | 0.9 | 0.91 | 1.66 | FAIL |
| FF-IT | SMDB-FF | IT room | 4C × 4 mm² | 20 | 29.3 | 0.95 | 1.29 | 2.04 | FAIL |

| Circuit | From | To | Cable | Length (m) | Ib (A) | PF | ΔV cable (%) | ΔV total (%) | Result |
|---|---|---|---|---|---|---|---|---|---|
| FF-LAB | SMDB-FF | Lab | 4C × 16 mm² | 30 | 57.3 | 0.85 | 0.87 | 1.62 | FAIL |
| FF-LTG | SMDB-FF | Corridor lighting | 2C × 6 mm² | 35 | 18.5 | 0.9 | 1.82 | 2.57 | FAIL |
| MCC-WP | MCC-1 | Water pump | 4C × 50 mm² | 20 | 121.3 | 0.86 | 0.45 | 1.03 | FAIL |
| MCC-FP | MCC-1 | Fire pump | 4C × 35 mm² | 25 | 89 | 0.86 | 0.54 | 1.12 | FAIL |
| INC-DBGF1 | SMDB-GF | DB-GF1 | 4C × 35 mm² | 30 | 60.4 | 0.9 | 0.45 | 0.92 | FAIL |

## B.3 DB and cable schedules — Cable schedule

**Table 35**

| Cable tag | From | To | Ref | Type | Cores × size | CPC (mm²) | Length (m) | Tray route | Ib (A) | Breaker In (A) | Iz (A) | Vd cable (%) | Vd total (%) | Status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| C-INC-GF | MDB-1 | SMDB-GF | 1 | Cu/XLPE/SWA/PVC (armoured) | 4C × 300 mm² | 150 | 35 | | 327 | 500 | 525 | 0.46 | 0.46 | WARNING |
| C-INC-FF | MDB-1 | SMDB-FF | 6 | Cu/XLPE/SWA/PVC (armoured) | 4C × 95 mm² | 50 | 55 | | 135 | 200 | 266 | 0.75 | 0.75 | FAIL |
| C-INC-MCC | MDB-1 | MCC-1 | 4 | Cu/XLPE/SWA/PVC (armoured) | 4C × 150 mm² | 95 | 40 | | 210 | 315 | 342 | 0.58 | 0.58 | FAIL |
| C-DB-EV | MDB-1 | EV charging – parking | 12 | Cu/XLPE/SWA/PVC (armoured) | 4C × 10 mm² | 10 | 60 | | 44 | 63 | 70 | 2.38 | 2.38 | FAIL |
| C-DB-PV | MDB-1 | Solar PV inverter | 10 | Cu/XLPE/SWA/PVC (armoured) | 4C × 25 mm² | 16 | 30 | | 70 | 100 | 120 | 0.76 | 0.76 | FAIL |
| C-GF-LTG | SMDB-GF | Lighting | 11 | Cu/XLPE/SWA/PVC (armoured) | 4C × 16 mm² | 16 | 30 | | 63 | 80 | 91 | 1 | 1.47 | FAIL |
| C-GF-SKT | SMDB-GF | Sockets | 14 | Cu/XLPE/SWA/PVC (armoured) | 4C × 4 mm² | 4 | 35 | | 27 | 40 | 41 | 1.91 | 2.37 | FAIL |
| C-GF-HVAC | SMDB-GF | HVAC | 6 | Cu/XLPE/SWA/PVC (armoured) | 4C × 95 mm² | 50 | 40 | | 177 | 250 | 266 | 0.7 | 1.16 | FAIL |
| C-INC-DBGF1 | SMDB-GF | DB-GF1 | 9 | Cu/XLPE/SWA/PVC (armoured) | 4C × 35 mm² | 16 | 30 | | 60 | 100 | 146 | 0.45 | 0.92 | FAIL |
| C-DB-GF1-Y1 | DB-GF1 | Majlis lighting | 28 | Cu/XLPE/SWA/PVC (armoured) | 2C × 2.5 mm² | 2.5 | 18 | | 5 | 6 | 31 | 0.56 | 1.48 | FAIL |
| C-DB-GF1-R1 | DB-GF1 | Living lighting | 28 | Cu/XLPE/SWA/PVC (armoured) | 2C × 2.5 mm² | 2.5 | 20 | | 6 | 10 | 31 | 0.75 | 1.67 | FAIL |

| Cable tag | From | To | Ref | Type | Cores × size | CPC (mm²) | Length (m) | Tray route | Ib (A) | Breaker In (A) | Iz (A) | Vd cable (%) | Vd total (%) | Status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| C-DB-GF1-Y2 | DB-GF1 | Kitchen lighting | 28 | Cu/XLPE/SWA/PVC (armoured) | 2C × 2.5 mm² | 2.5 | 22 | | 4 | 6 | 31 | 0.58 | 1.49 | FAIL |
| C-DB-GF1-B6 | DB-GF1 | Majlis sockets | 26 | Cu/XLPE/SWA/PVC (armoured) | 2C × 4 mm² | 4 | 18 | | 7 | 10 | 41 | 0.52 | 1.44 | FAIL |
| C-DB-GF1-Y5 | DB-GF1 | Living sockets | 26 | Cu/XLPE/SWA/PVC (armoured) | 2C × 4 mm² | 4 | 20 | | 9 | 16 | 41 | 0.78 | 1.7 | FAIL |
| C-DB-GF1-B7 | DB-GF1 | Kitchen sockets | 26 | Cu/XLPE/SWA/PVC (armoured) | 2C × 4 mm² | 4 | 22 | | 6 | 10 | 41 | 0.53 | 1.45 | FAIL |
| C-DB-GF1-R3 | DB-GF1 | Cooker | 25 | Cu/XLPE/SWA/PVC (armoured) | 2C × 6 mm² | 6 | 24 | | 28 | 40 | 52 | 1.88 | 2.79 | FAIL |
| C-DB-GF1-B3 | DB-GF1 | Kitchen water heater | 26 | Cu/XLPE/SWA/PVC (armoured) | 2C × 4 mm² | 4 | 24 | | 14 | 20 | 41 | 1.4 | 2.32 | FAIL |
| C-DB-GF1-R6 | DB-GF1 | Maid room | 26 | Cu/XLPE/SWA/PVC (armoured) | 2C × 4 mm² | 4 | 28 | | 5 | 6 | 41 | 0.59 | 1.51 | FAIL |
| C-DB-GF1-B4 | DB-GF1 | Majlis split A/C | 26 | Cu/XLPE/SWA/PVC (armoured) | 2C × 4 mm² | 4 | 18 | | 12 | 16 | 41 | 0.87 | 1.79 | FAIL |
| C-DB-GF1-Y4 | DB-GF1 | Living split A/C | 26 | Cu/XLPE/SWA/PVC (armoured) | 2C × 4 mm² | 4 | 20 | | 12 | 16 | 41 | 0.97 | 1.89 | FAIL |
| C-DB-GF1-B5 | DB-GF1 | Dining split A/C | 26 | Cu/XLPE/SWA/PVC (armoured) | 2C × 4 mm² | 4 | 22 | | 12 | 16 | 41 | 1.07 | 1.99 | FAIL |
| C-DB-GF1-R5 | DB-GF1 | Guest bedroom | 26 | Cu/XLPE/SWA/PVC (armoured) | 2C × 4 mm² | 4 | 26 | | 7 | 10 | 41 | 0.75 | 1.67 | FAIL |
| C-DB-GF1-Y8 | DB-GF1 | Guest bathroom | 26 | Cu/XLPE/SWA/PVC (armoured) | 2C × 4 mm² | 4 | 27 | | 1 | 6 | 41 | 0.14 | 1.05 | FAIL |
| C-DB-GF1-Y3 | DB-GF1 | Guest bath water heater | 26 | Cu/XLPE/SWA/PVC (armoured) | 2C × 4 mm² | 4 | 27 | | 14 | 20 | 41 | 1.57 | 2.49 | FAIL |
| C-DB-GF1-R4 | DB-GF1 | Guest bedroom split A/C | 26 | Cu/XLPE/SWA/PVC (armoured) | 2C × 4 mm² | 4 | 26 | | 12 | 16 | 41 | 1.26 | 2.18 | FAIL |
| C-DB-GF1-B2 | DB-GF1 | Dining lighting | 28 | Cu/XLPE/SWA/PVC (armoured) | 2C × 2.5 mm² | 2.5 | 22 | | 3 | 6 | 31 | 0.41 | 1.33 | FAIL |
| C-DB-GF1-Y7 | DB-GF1 | Dining sockets | 26 | Cu/XLPE/SWA/PVC (armoured) | 2C × 4 mm² | 4 | 22 | | 5 | 6 | 41 | 0.43 | 1.35 | FAIL |

| Cable tag | From | To | Ref | Type | Cores × size | CPC (mm²) | Length (m) | Tray route | Ib (A) | Breaker In (A) | Iz (A) | Vd cable (%) | Vd total (%) | Status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| C-DB-GF1-B1 | DB-GF1 | External lighting | 28 | Cu/XLPE/SWA/PVC (armoured) | 2C × 2.5 mm² | 2.5 | 35 | | 5 | 6 | 31 | 1.09 | 2.01 | FAIL |
| C-DB-GF1-R7 | DB-GF1 | Water pump | 26 | Cu/XLPE/SWA/PVC (armoured) | 2C × 4 mm² | 4 | 30 | | 3 | 6 | 41 | 0.44 | 1.36 | FAIL |
| C-DB-GF1-Y6 | DB-GF1 | Laundry | 26 | Cu/XLPE/SWA/PVC (armoured) | 2C × 4 mm² | 4 | 25 | | 9 | 16 | 41 | 0.97 | 1.89 | FAIL |
| C-FF-OFF | SMDB-FF | Office | 12 | Cu/XLPE/SWA/PVC (armoured) | 4C × 10 mm² | 10 | 25 | | 43 | 63 | 70 | 0.91 | 1.66 | FAIL |
| C-FF-IT | SMDB-FF | IT room | 14 | Cu/XLPE/SWA/PVC (armoured) | 4C × 4 mm² | 4 | 20 | | 29 | 40 | 41 | 1.29 | 2.04 | FAIL |
| C-FF-LAB | SMDB-FF | Lab | 11 | Cu/XLPE/SWA/PVC (armoured) | 4C × 16 mm² | 16 | 30 | | 57 | 80 | 91 | 0.87 | 1.62 | FAIL |
| C-FF-LTG | SMDB-FF | Corridor lighting | 25 | Cu/XLPE/SWA/PVC (armoured) | 2C × 6 mm² | 6 | 35 | | 19 | 25 | 52 | 1.82 | 2.57 | FAIL |
| C-MCC-WP | MCC-1 | Water pump | 8 | Cu/XLPE/SWA/PVC (armoured) | 4C × 50 mm² | 25 | 20 | | 121 | 160 | 174 | 0.45 | 1.03 | FAIL |
| C-MCC-FP | MCC-1 | Fire pump | 9 * | Fire-rated, BS 8491 (120 min, large cables) | 4C × 35 mm² | 16 | 25 | | 89 | 125 | 146 | 0.54 | 1.12 | FAIL |

## APPENDIX C — SHORT-CIRCUIT AND EARTH FAULT CALCULATIONS

### C.1 Short circuit study — Breaker breaking capacity

**Table 36**

| Circuit | Board | To | Breaker | Ik″ at breaker (kA) | Icu (kA) | Margin | Result |
|---|---|---|---|---|---|---|---|
| INC-GF | MDB-1 | SMDB-GF | 500 A MCCB | 29.2 | 36 | 23 % | PASS |
| INC-FF | MDB-1 | SMDB-FF | 200 A MCCB | 29.2 | 36 | 23 % | PASS |
| INC-MCC | MDB-1 | MCC-1 | 315 A MCCB | 29.2 | 36 | 23 % | PASS |
| DB-EV | MDB-1 | EV charging – parking | 63 A C | 29.2 | 36 | 23 % | PASS |
| DB-PV | MDB-1 | Solar PV inverter | 100 A MCCB | 29.2 | 36 | 23 % | PASS |
| GF-LTG | SMDB-GF | Lighting | 80 A MCCB | 21.8 | 25 | 15 % | PASS |
| GF-SKT | SMDB-GF | Sockets | 40 A C | 21.8 | 25 | 15 % | PASS |
| GF-HVAC | SMDB-GF | HVAC | 250 A MCCB | 21.8 | 25 | 15 % | PASS |
| FF-OFF | SMDB-FF | Office | 63 A C | 14.3 | 25 | 75 % | PASS |
| FF-IT | SMDB-FF | IT room | 40 A C | 14.3 | 25 | 75 % | PASS |
| FF-LAB | SMDB-FF | Lab | 80 A MCCB | 14.3 | 25 | 75 % | PASS |
| FF-LTG | SMDB-FF | Corridor lighting | 25 A C | 14.3 | 25 | 75 % | PASS |
| MCC-WP | MCC-1 | Water pump | 160 A MCCB | 19 | 25 | 32 % | PASS |
| MCC-FP | MCC-1 | Fire pump | 125 A MCCB | 19 | 25 | 32 % | PASS |
| INC-DBGF1 | SMDB-GF | DB-GF1 | 100 A MCCB | 21.8 | 25 | 15 % | PASS |
| DB-GF1-Y1 | DB-GF1 | Majlis lighting | 6 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-R1 | DB-GF1 | Living lighting | 10 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-Y2 | DB-GF1 | Kitchen lighting | 6 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-B6 | DB-GF1 | Majlis sockets | 10 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-Y5 | DB-GF1 | Living sockets | 16 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-B7 | DB-GF1 | Kitchen sockets | 10 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-R3 | DB-GF1 | Cooker | 40 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-B3 | DB-GF1 | Kitchen water heater | 20 A C | 10.7 | 15 | 41 % | PASS |

| Circuit | Board | To | Breaker | Ik″ at breaker (kA) | Icu (kA) | Margin | Result |
|---|---|---|---|---|---|---|---|
| DB-GF1-R6 | DB-GF1 | Maid room | 6 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-B4 | DB-GF1 | Majlis split A/C | 16 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-Y4 | DB-GF1 | Living split A/C | 16 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-B5 | DB-GF1 | Dining split A/C | 16 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-R5 | DB-GF1 | Guest bedroom | 10 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-Y8 | DB-GF1 | Guest bathroom | 6 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-Y3 | DB-GF1 | Guest bath water heater | 20 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-R4 | DB-GF1 | Guest bedroom split A/C | 16 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-B2 | DB-GF1 | Dining lighting | 6 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-Y7 | DB-GF1 | Dining sockets | 6 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-B1 | DB-GF1 | External lighting | 6 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-R7 | DB-GF1 | Water pump | 6 A C | 10.7 | 15 | 41 % | PASS |
| DB-GF1-Y6 | DB-GF1 | Laundry | 16 A C | 10.7 | 15 | 41 % | PASS |

## C.2 Short circuit study — Fault at the end of each cable

**Table 37**

| Circuit | From | To | Cable | Length (m) | Ik at cable end (kA) |
|---|---|---|---|---|---|
| INC-GF | MDB-1 | SMDB-GF | 4C × 300 mm² | 35 | 21.81 |
| INC-FF | MDB-1 | SMDB-FF | 4C × 95 mm² | 55 | 14.3 |
| INC-MCC | MDB-1 | MCC-1 | 4C × 150 mm² | 40 | 19.01 |
| DB-EV | MDB-1 | EV charging – parking | 4C × 10 mm² | 60 | 2.24 |
| DB-PV | MDB-1 | Solar PV inverter | 4C × 25 mm² | 30 | 9.7 |
| GF-LTG | SMDB-GF | Lighting | 4C × 16 mm² | 30 | 6.2 |
| GF-SKT | SMDB-GF | Sockets | 4C × 4 mm² | 35 | 1.52 |
| GF-HVAC | SMDB-GF | HVAC | 4C × 95 mm² | 40 | 13.94 |
| FF-OFF | SMDB-FF | Office | 4C × 10 mm² | 25 | 4.2 |
| FF-IT | SMDB-FF | IT room | 4C × 4 mm² | 20 | 2.38 |

| Circuit | From | To | Cable | Length (m) | Ik at cable end (kA) |
|---|---|---|---|---|---|
| FF-LAB | SMDB-FF | Lab | 4C × 16 mm² | 30 | 5.11 |
| FF-LTG | SMDB-FF | Corridor lighting | 2C × 6 mm² | 35 | 1.1 |
| MCC-WP | MCC-1 | Water pump | 4C × 50 mm² | 20 | 12.95 |
| MCC-FP | MCC-1 | Fire pump | 4C × 35 mm² | 25 | 10.53 |
| INC-DBGF1 | SMDB-GF | DB-GF1 | 4C × 35 mm² | 30 | 10.66 |
| DB-GF1-Y1 | DB-GF1 | Majlis lighting | 2C × 2.5 mm² | 18 | 0.88 |
| DB-GF1-R1 | DB-GF1 | Living lighting | 2C × 2.5 mm² | 20 | 0.8 |
| DB-GF1-Y2 | DB-GF1 | Kitchen lighting | 2C × 2.5 mm² | 22 | 0.73 |
| DB-GF1-B6 | DB-GF1 | Majlis sockets | 2C × 4 mm² | 18 | 1.35 |
| DB-GF1-Y5 | DB-GF1 | Living sockets | 2C × 4 mm² | 20 | 1.23 |
| DB-GF1-B7 | DB-GF1 | Kitchen sockets | 2C × 4 mm² | 22 | 1.13 |
| DB-GF1-R3 | DB-GF1 | Cooker | 2C × 6 mm² | 24 | 1.5 |
| DB-GF1-B3 | DB-GF1 | Kitchen water heater | 2C × 4 mm² | 24 | 1.04 |
| DB-GF1-R6 | DB-GF1 | Maid room | 2C × 4 mm² | 28 | 0.9 |
| DB-GF1-B4 | DB-GF1 | Majlis split A/C | 2C × 4 mm² | 18 | 1.35 |
| DB-GF1-Y4 | DB-GF1 | Living split A/C | 2C × 4 mm² | 20 | 1.23 |
| DB-GF1-B5 | DB-GF1 | Dining split A/C | 2C × 4 mm² | 22 | 1.13 |
| DB-GF1-R5 | DB-GF1 | Guest bedroom | 2C × 4 mm² | 26 | 0.97 |
| DB-GF1-Y8 | DB-GF1 | Guest bathroom | 2C × 4 mm² | 27 | 0.94 |
| DB-GF1-Y3 | DB-GF1 | Guest bath water heater | 2C × 4 mm² | 27 | 0.94 |
| DB-GF1-R4 | DB-GF1 | Guest bedroom split A/C | 2C × 4 mm² | 26 | 0.97 |
| DB-GF1-B2 | DB-GF1 | Dining lighting | 2C × 2.5 mm² | 22 | 0.73 |
| DB-GF1-Y7 | DB-GF1 | Dining sockets | 2C × 4 mm² | 22 | 1.13 |
| DB-GF1-B1 | DB-GF1 | External lighting | 2C × 2.5 mm² | 35 | 0.47 |
| DB-GF1-R7 | DB-GF1 | Water pump | 2C × 4 mm² | 30 | 0.85 |
| DB-GF1-Y6 | DB-GF1 | Laundry | 2C × 4 mm² | 25 | 1 |

## C.3 Earth fault loop impedance and disconnection — Earth fault loop

**Table 38**

| Circuit | Board | Breaker | CPC (mm²) | Zs (Ω) | Max Zs (Ω) | If (A) | Ia (A) | Required (s) | Disconnection | CPC min, each (mm²) | Result |
|---|---|---|---|---|---|---|---|---|---|---|---|
| INC-GF | MDB-1 | 500 A MCCB | 150 | 0.0144 | 0.0455 | 15804 | 5000 | 5 | < 0.1 s | 34.9 | PASS |
| INC-FF | MDB-1 | 200 A MCCB | 50 | 0.0419 | 0.1138 | 5433 | 2000 | 5 | < 0.1 s | 12 | PASS |
| INC-MCC | MDB-1 | 315 A MCCB | 95 | 0.0204 | 0.0723 | 11150 | 3150 | 5 | < 0.1 s | 24.7 | PASS |
| DB-EV | MDB-1 | 63 A C | 10 | 0.2656 | 0.3613 | 857 | 630 | 0.4 | < 0.1 s | 1.9 | PASS |
| DB-PV | MDB-1 | 100 A MCCB | 16 | 0.0701 | 0.2276 | 3246 | 1000 | 5 | < 0.1 s | 7.2 | PASS |
| GF-LTG | SMDB-GF | 80 A MCCB | 16 | 0.0932 | 0.2845 | 2442 | 800 | 5 | < 0.1 s | 5.4 | PASS |
| GF-SKT | SMDB-GF | 40 A C | 4 | 0.3969 | 0.5691 | 573 | 400 | 0.4 | < 0.1 s | 1.3 | PASS |
| GF-HVAC | SMDB-GF | 250 A MCCB | 50 | 0.0398 | 0.091 | 5724 | 2500 | 5 | < 0.1 s | 12.7 | PASS |
| FF-OFF | SMDB-FF | 63 A C | 10 | 0.1505 | 0.3613 | 1512 | 630 | 0.4 | < 0.1 s | 3.3 | PASS |
| FF-IT | SMDB-FF | 40 A C | 4 | 0.2616 | 0.5691 | 870 | 400 | 0.4 | < 0.1 s | 1.9 | PASS |
| FF-LAB | SMDB-FF | 80 A MCCB | 16 | 0.1237 | 0.2845 | 1840 | 800 | 5 | < 0.1 s | 4.1 | PASS |
| FF-LTG | SMDB-FF | 25 A C | 6 | 0.2991 | 0.9105 | 761 | 250 | 0.4 | < 0.1 s | 1.7 | PASS |
| MCC-WP | MCC-1 | 160 A MCCB | 25 | 0.0455 | 0.1423 | 4997 | 1600 | 5 | < 0.1 s | 11.1 | PASS |
| MCC-FP | MCC-1 | 125 A MCCB | 16 | 0.0685 | 0.1821 | 3325 | 1250 | 5 | < 0.1 s | 7.4 | PASS |
| INC-DBGF1 | SMDB-GF | 100 A MCCB | 16 | 0.0709 | 0.2276 | 3209 | 1000 | 5 | < 0.1 s | 7.1 | PASS |
| DB-GF1-Y1 | DB-GF1 | 6 A C | 2.5 | 0.3901 | 455.2407 | 584 | 1 | 0.4 | < 0.1 s | 1.3 | PASS |
| DB-GF1-R1 | DB-GF1 | 10 A C | 2.5 | 0.4256 | 455.2407 | 535 | 1 | 0.4 | < 0.1 s | 1.2 | PASS |
| DB-GF1-Y2 | DB-GF1 | 6 A C | 2.5 | 0.4612 | 455.2407 | 494 | 1 | 0.4 | < 0.1 s | 1.1 | PASS |
| DB-GF1-B6 | DB-GF1 | 10 A C | 4 | 0.2693 | 1517.469 | 845 | 0 | 0.4 | < 0.1 s | 1.9 | PASS |
| DB-GF1-Y5 | DB-GF1 | 16 A C | 4 | 0.2914 | 1517.469 | 781 | 0 | 0.4 | < 0.1 s | 1.7 | PASS |
| DB-GF1-B7 | DB-GF1 | 10 A C | 4 | 0.3135 | 1517.469 | 726 | 0 | 0.4 | < 0.1 s | 1.6 | PASS |
| DB-GF1-R3 | DB-GF1 | 40 A C | 6 | 0.2476 | 1517.469 | 919 | 0 | 0.4 | < 0.1 s | 2 | PASS |
| DB-GF1-B3 | DB-GF1 | 20 A C | 4 | 0.3356 | 1517.469 | 678 | 0 | 0.4 | < 0.1 s | 1.5 | PASS |
| DB-GF1-R6 | DB-GF1 | 6 A C | 4 | 0.3798 | 1517.469 | 599 | 0 | 0.4 | < 0.1 s | 1.3 | PASS |
| DB-GF1-B4 | DB-GF1 | 16 A C | 4 | 0.2693 | 1517.469 | 845 | 0 | 0.4 | < 0.1 s | 1.9 | PASS |

| Circuit | Board | Breaker | CPC (mm²) | Zs (Ω) | Max Zs (Ω) | If (A) | Ia (A) | Required (s) | Disconnection | CPC min, each (mm²) | Result |
|---|---|---|---|---|---|---|---|---|---|---|---|
| DB-GF1-Y4 | DB-GF1 | 16 A C | 4 | 0.2914 | 1517.469 | 781 | 0 | 0.4 | < 0.1 s | 1.7 | PASS |
| DB-GF1-B5 | DB-GF1 | 16 A C | 4 | 0.3135 | 1517.469 | 726 | 0 | 0.4 | < 0.1 s | 1.6 | PASS |
| DB-GF1-R5 | DB-GF1 | 10 A C | 4 | 0.3577 | 1517.469 | 636 | 0 | 0.4 | < 0.1 s | 1.4 | PASS |
| DB-GF1-Y8 | DB-GF1 | 6 A C | 4 | 0.3687 | 1517.469 | 617 | 0 | 0.4 | < 0.1 s | 1.4 | PASS |
| DB-GF1-Y3 | DB-GF1 | 20 A C | 4 | 0.3687 | 1517.469 | 617 | 0 | 0.4 | < 0.1 s | 1.4 | PASS |
| DB-GF1-R4 | DB-GF1 | 16 A C | 4 | 0.3577 | 1517.469 | 636 | 0 | 0.4 | < 0.1 s | 1.4 | PASS |
| DB-GF1-B2 | DB-GF1 | 6 A C | 2.5 | 0.4612 | 455.2407 | 494 | 1 | 0.4 | < 0.1 s | 1.1 | PASS |
| DB-GF1-Y7 | DB-GF1 | 6 A C | 4 | 0.3135 | 1517.469 | 726 | 0 | 0.4 | < 0.1 s | 1.6 | PASS |
| DB-GF1-B1 | DB-GF1 | 6 A C | 2.5 | 0.6923 | 455.2407 | 329 | 1 | 0.4 | < 0.1 s | 0.7 | PASS |
| DB-GF1-R7 | DB-GF1 | 6 A C | 4 | 0.4019 | 1517.469 | 566 | 0 | 0.4 | < 0.1 s | 1.3 | PASS |
| DB-GF1-Y6 | DB-GF1 | 16 A C | 4 | 0.3466 | 1517.469 | 657 | 0 | 0.4 | < 0.1 s | 1.5 | PASS |

## APPENDIX D — EQUIPMENT DATA

### D.1 Transformer and standby generator sizing — Standby generator — loads

**Table 39**

| Board / circuit | Share | Demand (kW) | Reactive (kvar) |
|---|---|---|---|
| MCC-FP Fire pump (essential circuit) | 100 % | 55 | — |

### D.2 Boards

**Table 40**

| Board | Type | Rating (A) | Busbar | IP | Manufacturer | Model | Location |
|---|---|---|---|---|---|---|---|
| MDB-1 | MDB | 1600 | copper | IP42 | Schneider Electric | PrismaSeT P | Main building – ground floor |
| SMDB-GF | SMDB | 630 | copper | IP42 | Not defined | Not defined | Ground floor electrical room |
| SMDB-FF | SMDB | 250 | copper | IP42 | Not defined | Not defined | First floor electrical room |
| MCC-1 | MCC | 400 | copper | IP54 | Not defined | Not defined | Basement pump room |
| DB-GF1 | DB | 100 | copper | IP42 | Not defined | Not defined | Villa GF – store room |

### D.3 Transformers

**Table 41**

| Reference | Main board | Rating (kVA) | %Z | X/R | Vector group | Substation |
|---|---|---|---|---|---|---|
| MDB-1 | MDB-1 | 1000 | 5 | 5 (app default) | Dyn11 (app default) | Not defined |

# APPENDIX E — REFERENCE DOCUMENTS

## E.1 Calculation basis

**Table 42**

| Standard | Used for |
|---|---|
| IEC 60909-0 | Short-circuit currents (simplified: c factor, infinite MV network) |
| IEC 60364-5-52 | Cable current-carrying capacity, derating and voltage drop |
| IEC 60364-4-43 | Overload protection: Ib ≤ In ≤ Iz |
| IEC 60364-4-41 | Automatic disconnection: earth fault loop impedance and disconnection times |
| IEC 60364-5-54 | Protective conductor size (Table 54.2 / adiabatic) |
| IEC 60898-1 / IEC 60947-2 | Circuit-breaker characteristics and breaking capacity |
| IEC 60831 | Capacitor bank rating and switching device |
| IEC 61439-6 | Busbar trunking ratings |